

Two-step successor beliefs

Where a branching maze forks three-to-four ways, a dense predictor collapses onto a single mode; best-of-K re-opens every branch.

GROUND TRUTH

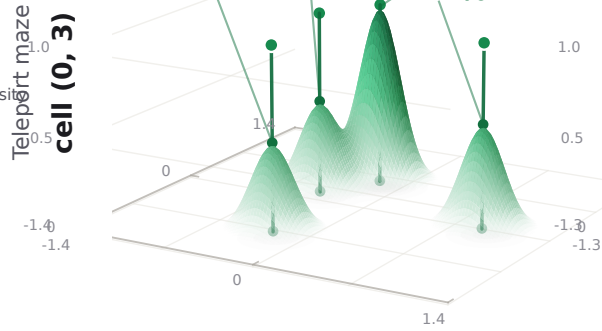
true 2-step successor law

19.8%

39.2%

18.8%

22.2%

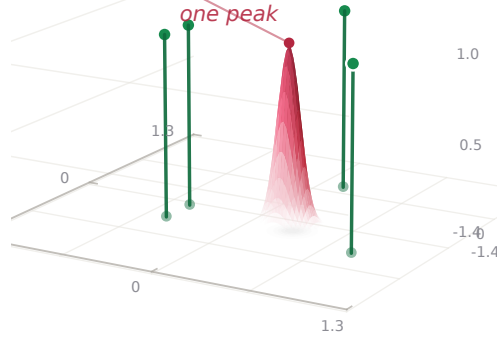


DENSE PREDICTOR

belief collapses to one mode

100.0%

one peak



BEST-OF-K · OURS

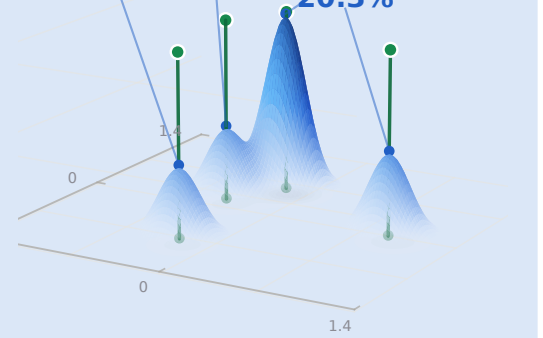
belief re-opens the full fan-out

17.8%

44.4%

17.5%

20.3%

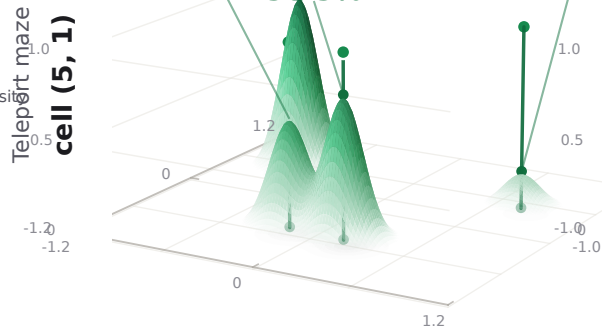


39.1%

7.5%

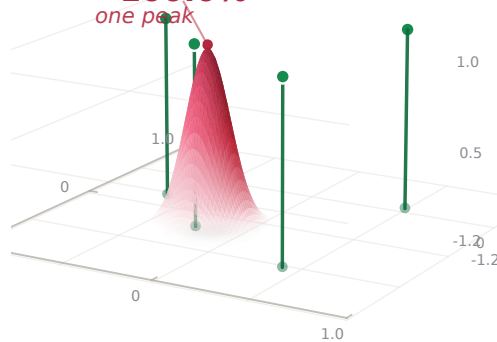
23.1%

30.3%



100.0%

one peak

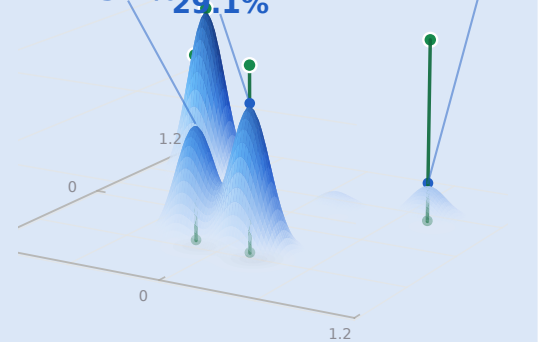


36.4%

7.3%

23.1%

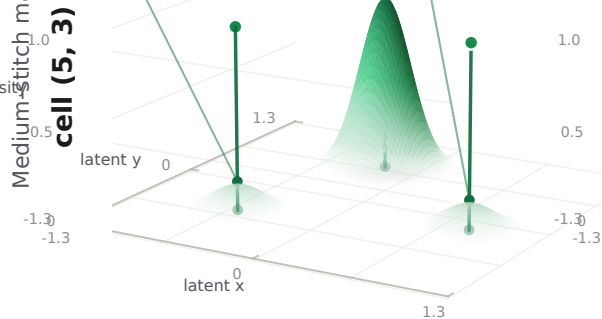
29.1%



77.2%

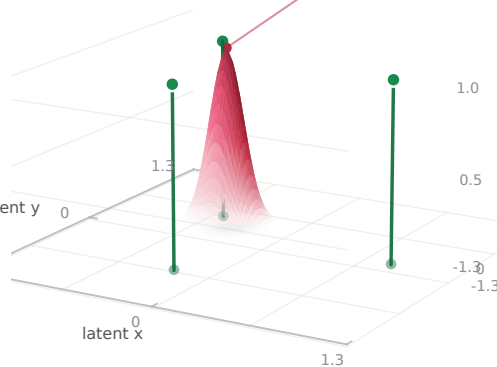
11.3%

11.6%



100.0%

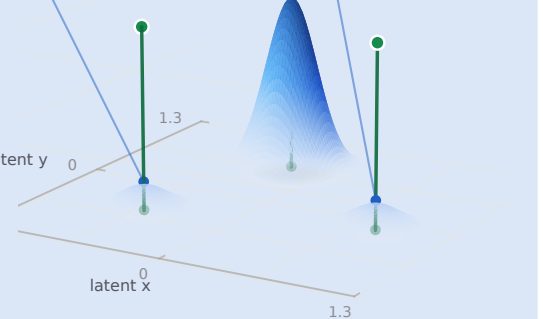
one peak



77.4%

11.2%

11.4%



Green pins = true successors. Dense puts 100% on a single mode; best-of-K restores a peak under every branch.